

Embedded System Design and Modeling

II. Modeling of continuous systems

Actor model of systems

A system can be decomposed into interconnected building blocks, called “actors”

- ▶ Each actor has:
 - ▶ 0, 1 or more input ports
 - ▶ 0, 1 or more output ports
 - ▶ an internal computation / function / what it does
- ▶ Connections carry signals between blocks

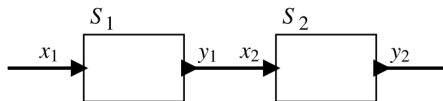


Figure 1: Actor model of systems¹

¹(Image from Lee&Seshia 2017)

Actor dynamics

$x(t)$

$x'(t)$

$x''(t)$



How to describe what a component does?

- ▶ Continuous dynamics
- ▶ Discrete dynamics



Similar to an ancient philosophy debate: - Heraclitus: everything is in flux (continuous change), stationarity is an illusion - Parmenides: everything is static (discrete change), movement is an illusion

Continuous dynamics

- ▶ **Dynamic system** = system whose state evolves in time
- ▶ **Continuous dynamics** = the state is described by continuous functions. We consider systems which are governed by **differential equations**
- ▶ The differential equations involve the unknown function $x(t)$ and its derivatives.
- ▶ Example: mechanical, electrical physical processes
 - ▶ governed by mechanical / electrical differential equations
 - ▶ example:

$$\rightarrow \boxed{m_1 \ddot{x}(t) + K(x(t) - x_0) = 0}$$
- ▶ Every electrical/mechanical component defines a certain relation between the unknowns



Electrical systems

Electrical systems:

$$u(t), i(t)$$

- ▶ Unknown functions = voltage + current in all branches
- ▶ Electrical (ideal) elements:
 - ▶ resistance: $u(t) = R \cdot i(t)$
 - ▶ capacitance: $i(t) = C \cdot \frac{d}{dt} u(t)$
 - ▶ etc.
- ▶ One big system of differential equations (SCS course, basically)
 - ▶ Kirchhoff equations $\Leftrightarrow$ equations between currents and voltages $\Leftrightarrow$ linear differential equation system
- ▶ ~~Example: an RC system (solve at blackboard)~~

Mechanical systems

Mechanical systems:

- ▶ Unknown functions = coordinates $x(t)$, $y(t)$, $z(t)$
 - ▶ velocities = derivatives of the positions
 - ▶ acceleration = derivative of velocity = second derivative of position
 - ▶ (forces: $F = m \cdot a = m \cdot \frac{d^2}{dt^2}x(t)$)
- ▶ Mechanical (ideal) elements:
 - ▶ (Consider just a single dimension $x(t)$, which is simpler)
 - ▶ inertial force: $F = m \cdot a = m \cdot \frac{d^2}{dt^2}x(t)$
 - ▶ friction force:
 - ▶ sliding friction: $F_f = -\mu N = -\mu \cdot m \cdot g$
 - ▶ viscous friction: $\vec{F}_v = -C_v \cdot \vec{v} = -C_v \cdot \frac{d}{dt}x(t)$
 - ▶ etc. . .

Mechanical systems

- ▶ Mechanical elements are described by linear differential equations, just like electrical ones
 - ▶ they are just idealizations, physical processes can be highly nonlinear (more complex)
 - ▶ but wait, so are electrical devices actually, and this hasn't stopped us. . .
- ▶ Example: oscillations after releasing a mass-spring system
 - ▶ (solve at blackboard)

Equivalence spring = LC circuit

- ▶ A loaded spring oscillates (without any friction) according to the equation:

$$-\frac{1}{m}kx = \frac{d^2x}{dt^2}$$

$$\ddot{x}(t) + \frac{k}{m} \cdot x(t) = 0$$

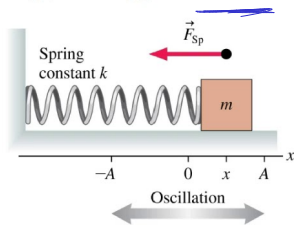
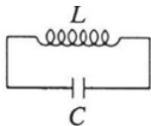


Figure 2: Spring oscillations

- ▶ image from <https://www.youtube.com/watch?v=M2m0ALqgcnQ>

Equivalence spring = LC circuit

- ▶ An LC circuit oscillates (without resistive losses) according to the equation:



Equation of motion for charge q
(no damping or resistance R):

$$L(d^2q/dt^2) + (q/C) = 0 \quad (1)$$

Electrical Oscillator

Figure 3: LC oscillations

$$L \cdot q''(t) + \frac{1}{C} \cdot q(t) = 0$$

(=)

$$q''(t) + \frac{1}{LC} q(t) = 0$$

- ▶ image from
<https://www.rfwireless-world.com/Terminology/Mechanical-Oscillator-vs-Electrical-Oscillator.html>

Equivalence spring = LC circuit

- ▶ Notice the similarities
- ▶ Same linear differential equation:

$$\frac{d^2}{dt^2}f(t) + A \cdot f(t) = 0$$

- ▶ Same solution
 - ▶ $f(t) = \text{sinusoidal}$ (why sinusoidal?)
- ▶ Many other types of continuous systems can be described using linear differential equations

Electrical - mechanical analogies

- ▶ Multiple ways to define analogies between electrical and mechanical characteristics
- ▶ Here is the one we will use from now on:

Electr.	Mech. (linear)	Mech. (rotational)
Current [A]	Force [N]	Tor <u>que</u> ("cuplu") [N.m]
Voltage [V]	Acc Speed [m/s]	Angu <u>lar</u> speed [rad/s]



Mechanics: linear vs rotational

- ▶ Note: there are different quantities for **linear** vs **rotational** movements
 - ▶ **Force** in linear movement $\equiv$ **Torque** (cuplu) in rotational movement
 - ▶ Linear speed in linear movement $\equiv$ Angular speed in rotational movement

Simple model of a DC motor

- ▶ Example of continuous system modeling: model of a DC motor
- ▶ Motor: gateway between the two electrical and mechanical domains
 - ▶ converts electric energy to mechanical energy, and vice-versa
- ▶ (Simple) model of a DC motor:

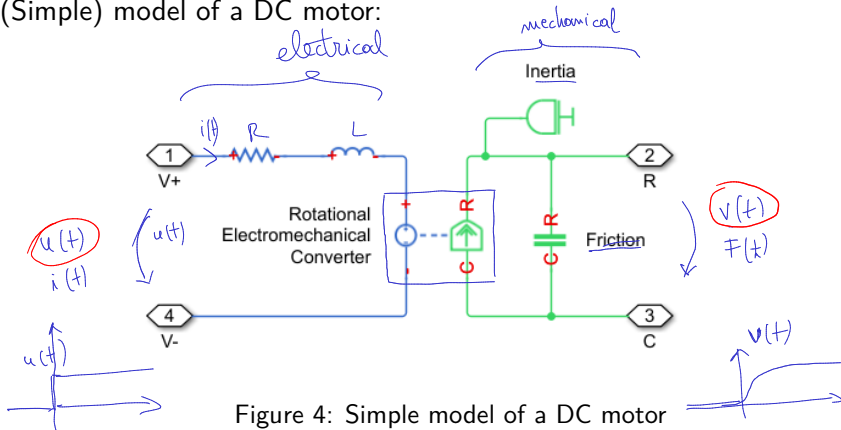


Figure 4: Simple model of a DC motor

DC motor model: electrical side

Electrical side of the DC motor model:

- ▶ Resistance: models the resistance of the windings

$$\underline{u(t)} = R \cdot i(t)$$

- ▶ Inductance: models the inductive behavior of the windings

$$u(t) = L \cdot \frac{d}{dt} i(t)$$

- ▶ Controlled voltage source:

- ▶ Voltage (“^{EMF}back electro-magnetic force voltage”) is proportional to motor angular speed $S(t)$ on the mechanical side (think of a dynamo)

$$\underline{u(t)} = K_e \cdot \underline{S(t)}$$

DC motor model: mechanical side

Mechanical circuit of the DC motor model (no load):

- ▶ Controlled force/torque source
 - ▶ Generates force/torque proportional to the current $i(t)$ on the electrical side

$$\underline{T} = K_t \cdot \underline{i(t)}$$

- ▶ Inertia: models the inertial force of the moving part of the motor
 - ▶ Generates force/torque proportional to acceleration (derivative of speed)
 - ▶ linear:

$$T_i = -m \cdot acceleration = -m \cdot \frac{d}{dt} S(t)$$

- ▶ rotational:

$$T_i = -J \cdot angular\ acceleration = -J \cdot \frac{d}{dt} S(t)$$

DC motor model: mechanical side

Mechanical circuit of the DC motor model (no load):

- ▶ Friction: models the (viscous) friction force of the moving part of the motor
 - ▶ Generates force/torque proportional to speed

$$T_f = -C_v \cdot \underline{S(t)}$$

- ▶ Inertia and Friction forces/torques oppose the movement of the motor, therefore they have minus sign

Laplace transform

- ▶ Both electrical and mechanical sides are described by linear differential equations
- ▶ The Laplace transform is a useful tool (remember SCS)
 - ▶ derivation = multiplication by s
 - ▶ integration = multiplication by $1/s$
 - ▶ ~~transform~~^{transfer} function $H(s) = \text{output}(s)/\text{input}(s)$
- ▶ Exercise: write the equations of all electrical and mechanical elements in Laplace transform

Full electrical model

- ▶ All the mechanical elements can be modeled in the electrical domain
 - ▶ since they are all just differential equations, basically
 - ▶ obtain a full model in the electrical domain only
- ▶ Next slides: find electrical correspondent to all mechanical elements

Model of the back EMF

- ▶ For zero load torque, the DC motor equations are:

$$u(t) = Ri(t) + L \frac{di(t)}{dt} + K_e \omega(t)$$

$$J \frac{d\omega(t)}{dt} + b\omega(t) = K_t i(t)$$

- ▶ Define the back-EMF node voltage as:

$$v_b(t) = K_e \omega(t)$$

- ▶ v_b is an electrical node voltage. The motor speed is $\omega = v_b/K_e$; they are not the same physical quantity.

Model of the inertial force

- ▶ Substitute $v_b = K_e \omega$ into the mechanical equation:

$$i(t) = \frac{J}{K_e K_t} \frac{dv_b(t)}{dt} + \frac{b}{K_e K_t} v_b(t)$$

- ▶ The first term is the current through an equivalent capacitor:

$$C_{eq} = \frac{J}{K_e K_t}$$

- ▶ Its initial voltage represents the initial motor speed:

$$v_b(0) = K_e \omega_0$$

- ▶ Inertia is represented once, by this single capacitor at the v_b node.

Model of the friction force

- ▶ The second term is the current through an equivalent resistor:

$$R_{eq} = \frac{K_e K_t}{b}$$

- ▶ Viscous friction is therefore represented by R_{eq} from the v_b node to the reference node.
- ▶ The capacitor and resistor currents add to the armature current:

$$i(t) = C_{eq} \frac{dv_b(t)}{dt} + \frac{v_b(t)}{R_{eq}}$$

Model of the sliding friction force

- ▶ There can also exist a sliding friction force = friction force which does not depend on speed, but is a constant
 - ▶ that's the friction force you likely encountered in high-school physics (“planul înclinat” etc.)
- ▶ Question: how is this force modeled in electrical domain?

Model of the sliding friction force

- ▶ Answer: a constant current source in parallel
 - ▶ constant current $\Leftrightarrow$ constant source
 - ▶ in parallel $\Leftrightarrow$ reduces the motor current

The full electrical model of a DC Motor

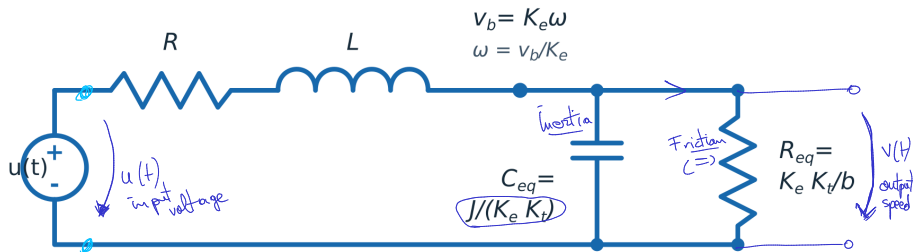


Figure 5: DC Motor electrical equivalent model

- ▶ R and L feed the back-EMF node $v_b = K_e \omega$.
- ▶ One shunt C_{eq} models inertia; R_{eq} models viscous friction.
- ▶ With one inductor and one capacitor, this model is **second order**.

Transfer function of a DC motor

- ▶ Under zero initial conditions and zero load torque:

- ▶ input = motor voltage $U(s)$
- ▶ back-EMF node voltage = $V_b(s)$
- ▶ output = motor speed $\Omega(s) = V_b(s)/K_e$

$$U(s) = (Ls + R)I(s) + V_b(s), \quad I(s) = \left(sC_{eq} + \frac{1}{R_{eq}} \right) V_b(s)$$

- ▶ Transfer function (second order, approximately first order)

$$\begin{aligned} H(s) &= \frac{\overset{\text{angular speed}}{\Omega(s)}}{\underset{\text{voltage}}{U(s)}} = \frac{1}{K_e} \frac{V_b(s)}{U(s)} \\ &= \frac{1}{K_e} \frac{1}{1 + (Ls + R) \left(sC_{eq} + \frac{1}{R_{eq}} \right)} \\ &= \frac{K_t}{(Ls + R)(Js + b) + K_e K_t} \end{aligned}$$

Transfer function of a DC motor

- ▶ Take home message:
 - ▶ Simple DC motor no-load model = a second order RLC model = approx a first-order RC model (ignoring L small)
 - ▶ The node voltage v_b is proportional to speed, with $\omega = v_b/K_e$
 - ▶ Behaves like an RC low-pass filter when the armature inductance is neglected
- ▶ Note: This is a no-load model (motor doesn't move an external load)
- ▶ What happens if motor has a load?
 - ▶ e.g. the motor drags/lifts a constant weight
 - ▶ i.e. like a crane lifting a big weight from the ground
- ▶ How to model the load?

Motor under load

- ▶ How to model the load?
- ▶ Like a constant force/torque opposing the motor force/torque
 - ▶ i.e. like a sliding friction force
 - ▶ i.e. like a current source in parallel, stealing lots of current
- ▶ In practice, the load force/torque may not be constant
 - ▶ depends on mechanical properties
 - ▶ e.g. lifting the hatch/liftgate (“portbagaj”) of a car: harder when lower, easier when higher

Simulink model

SimScope

Matlab

- ▶ Simulink has a DC motor model already integrated
- ▶ (You will use it in the lab (maybe))

What to use the model for?

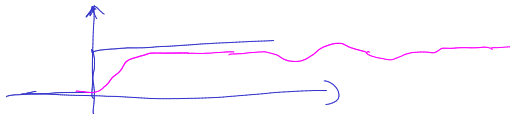
What to use the motor model for?

Simulate:

- ▶ how fast motor starts when supply is first applied
- ▶ what happens when supply fluctuates (e.g. PWM)
- ▶ what happens when motor parameters change (e.g. temperature rises, friction slows)
- ▶ what happens when load varies
- ▶ ...



Motor speed controller

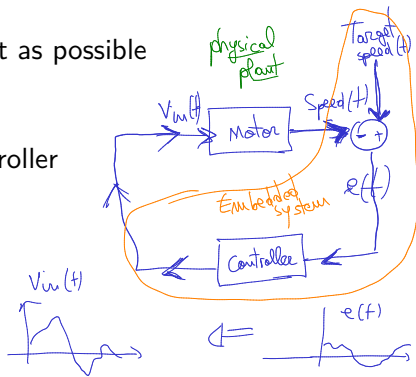


Basic problem: how to make sure motor speed stays exactly as desired:

- ▶ even if parameters vary
- ▶ even if load varies
- ▶ even if supply varies
- ▶ on power on, speed is reached as fast as possible

This is a job for a motor controller

- ▶ Today's special: the PID motor controller



Motor speed controller

This is a typical embedded system design problem:

- ▶ There is a physical process (the actual motor)
- ▶ We model its behavior (use a motor model)
- ▶ We want to control it
- ▶ We design a controller system which steers the process as we want

Motor controllers

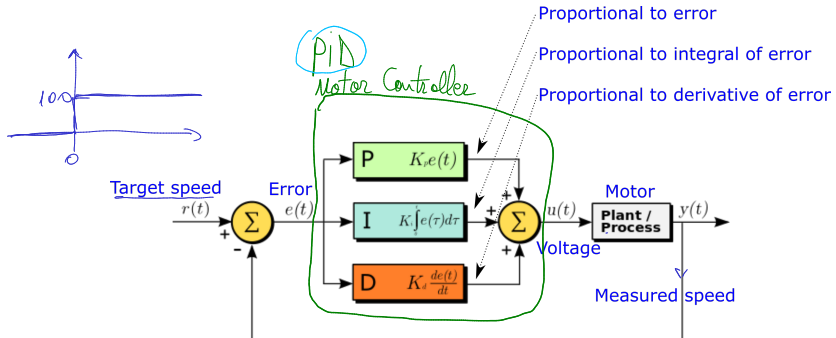


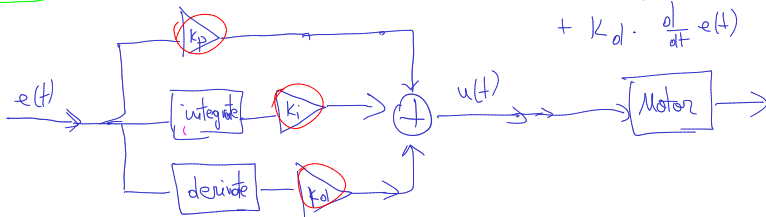
Figure 6: PID speed controller (image from Wikipedia)

- ▶ Negative feedback loop
- ▶ Can be used for any sort of process, not just motors
- ▶ Make output signal $y(t)$ follow the desired input $r(t)$

PID Controller

- ▶ PID controller = A common and simple solution
- ▶ Input = error signal = target speed - actual measured speed
- ▶ Output = Sum of three components:
 - ▶ Proportional: K_p * input
 - ▶ Integral: K_i * integral of input
 - ▶ Derivative: K_d * derivative of input

$$u(t) = K_p \cdot e(t) + K_i \int_0^t e(\tau) d\tau + K_d \cdot \frac{d}{dt} e(t)$$



PID Controller - P component

$$e(t) = \text{target} - \text{measured}$$

$$u(t) = K_p e(t) + K_i \cdot \int_0^t e(\tau) d\tau$$

- ▶ Intuitive role of the P component:
 - ▶ If actual speed < target $\Rightarrow$ increase motor voltage
 - ▶ If actual speed > target $\Rightarrow$ decrease motor voltage
- ▶ This is not enough:
 - ▶ Non-zero motor voltage requires non-zero speed error $\Rightarrow$ the motor doesn't actually reach the target speed
 - ▶ There is always a small systematic error ("steady-state error", "bias error")

PID controller - only P, systematic error

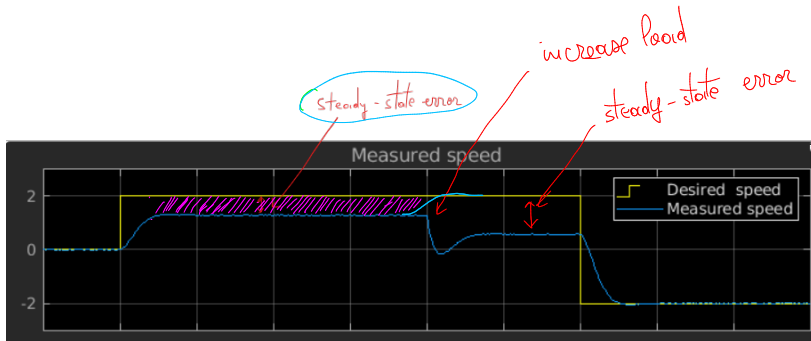


Figure 7: Systematic error for P-only controller

PID Controller - I component

- ▶ Intuitive role of the I component:
 - ▶ Eliminate the bias error of the P component, by slowly integrating the remaining error signal $\Rightarrow$ integral slowly increases over time $\Rightarrow$ motor voltage is pushed towards the correct value
 - ▶ Error signal cannot remain constant forever, because the integral would grow large $\Rightarrow$ force changes to the motor input

PID controller - P and I

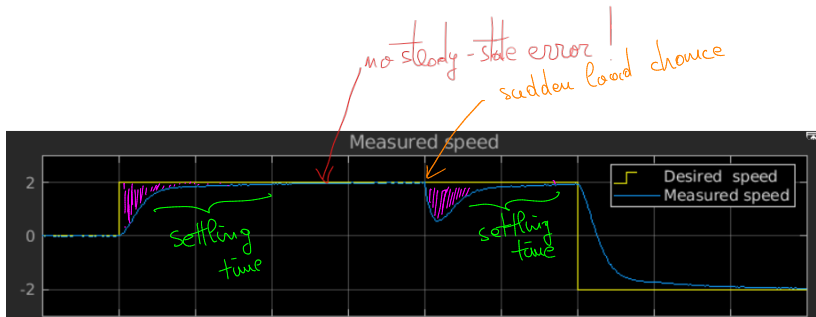


Figure 8: P and I components

PID Controller - D component

- ▶ The D component is harder to explain intuitively
- ▶ It makes the system react faster to changes in the error signal
 - ▶ for example, if the error signal suddenly increases, the derivative is positive $\Rightarrow$ the D component will increase the motor voltage even more than the P component alone would do
- ▶ Problems:
 - ▶ sensitive to noise (derivative of a noisy signal is very noisy)
 - ▶ possibly unstable

PID controller - P, I and D

faster response,
but more overshoot / oscillations!

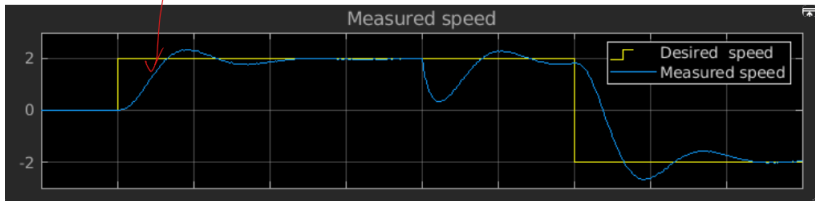


Figure 9: P, I and D components

PID tuning

- ▶ PID tuning: find P, I, D values for good behavior
 - ▶ Typical requirements:
 - ▶ stable system, overall
 - ▶ overshoot not larger than X%
 - ▶ fastest response in these conditions
- ▶ Find out more at the Vehicle Control Systems course (2nd semester, I think)